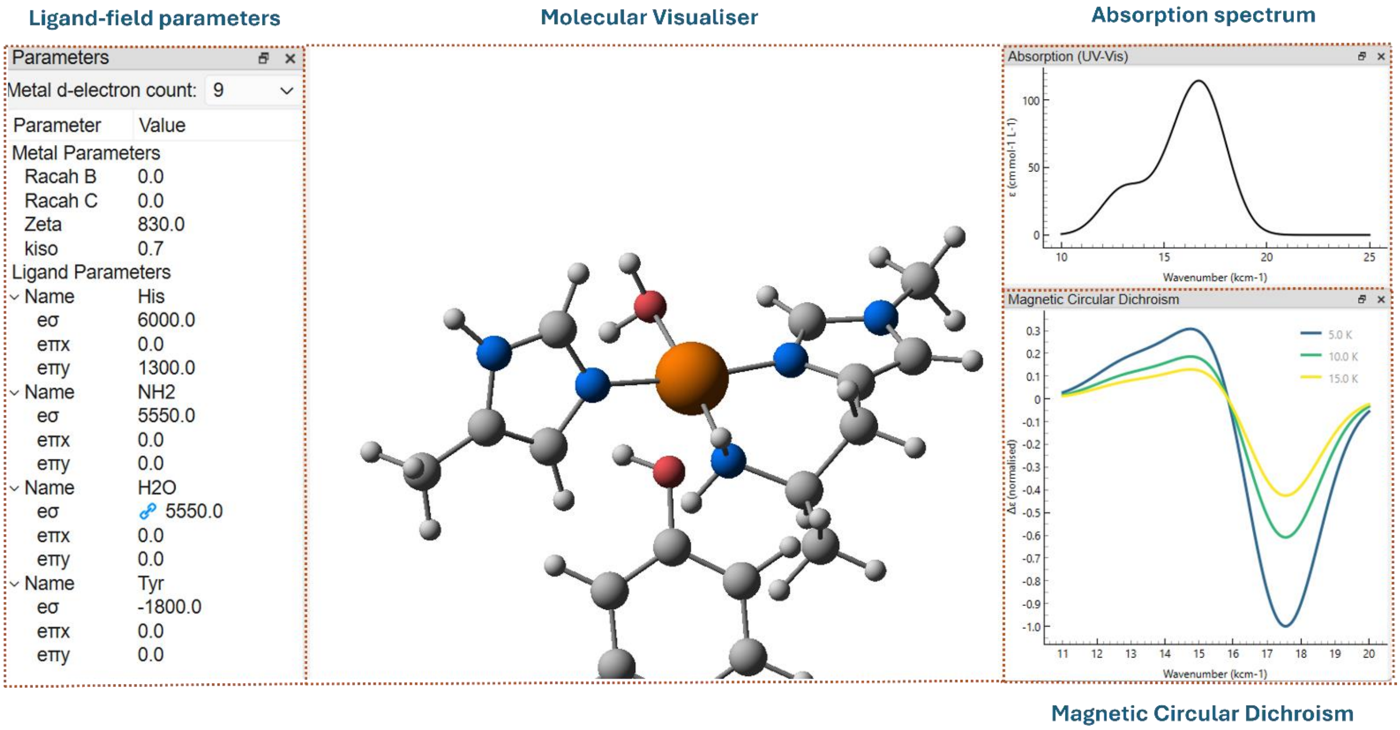
# *Kestrel* ‘quickstart’ guide




**Version 1.0 (16/04/2026)**

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https://chem-kestrel.github.io/

## Contributors:

- George Nunn – codebase and GUI development
- Benjamin S. Grace – GUI development and hyperfine module
- Paul H. Walton
- Martin A. Bates

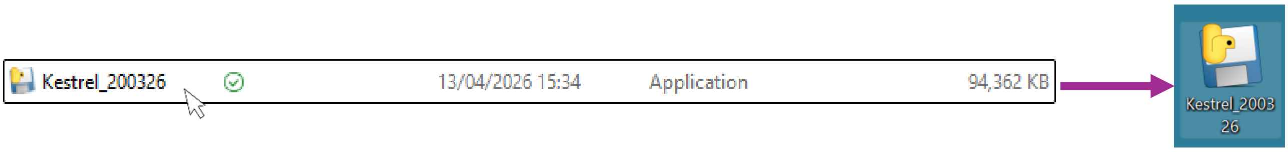
# Installation

Before running any calculations in *Kestrel*, the graphical user interface (GUI) or python codebase needs to be installed. This guide will focus on the installation and use of the GUI.

*Kestrel* will run on computers that support Windows 10 or 11, Linux or MacOS as a standalone program. No further prerequisites are required.

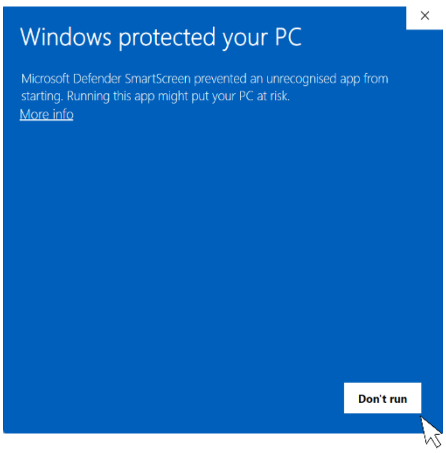
## Windows:

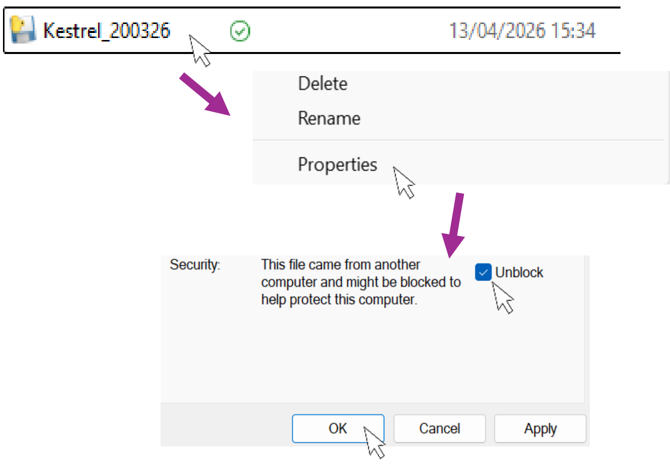
1) Download the latest version of *Kestrel* from our **website**. Open ‘File Explorer’ and navigate to your ‘downloads’ folder where you will see the *Kestrel* executable.
2) Optionally, this file can be dragged onto the desktop for ease of access.
3) Double click on the file/thumbnail to launch *Kestrel*.




### *Troubleshooting*

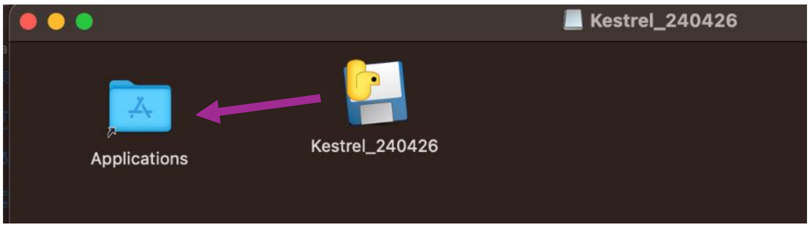
i) Depending on system security settings, the below message may be displayed.
ii) If this is the case, right click on the .exe file, select ‘properties’ and check the ‘Unblock’ box. Click ‘OK’ and *Kestrel* will run as expected.



## MacOS:

1) Download the latest version of *Kestrel* from our ***website***. Open 'Finder' and navigate to your 'downloads' folder where you will see the *Kestrel* .dmg file.
2) Double click on the .dmg file and drag the application ("*Kestrel.app*") to your 'Applications' folder (see below).
3) Double click on the thumbnail in the 'Applications' folder to launch *Kestrel*.

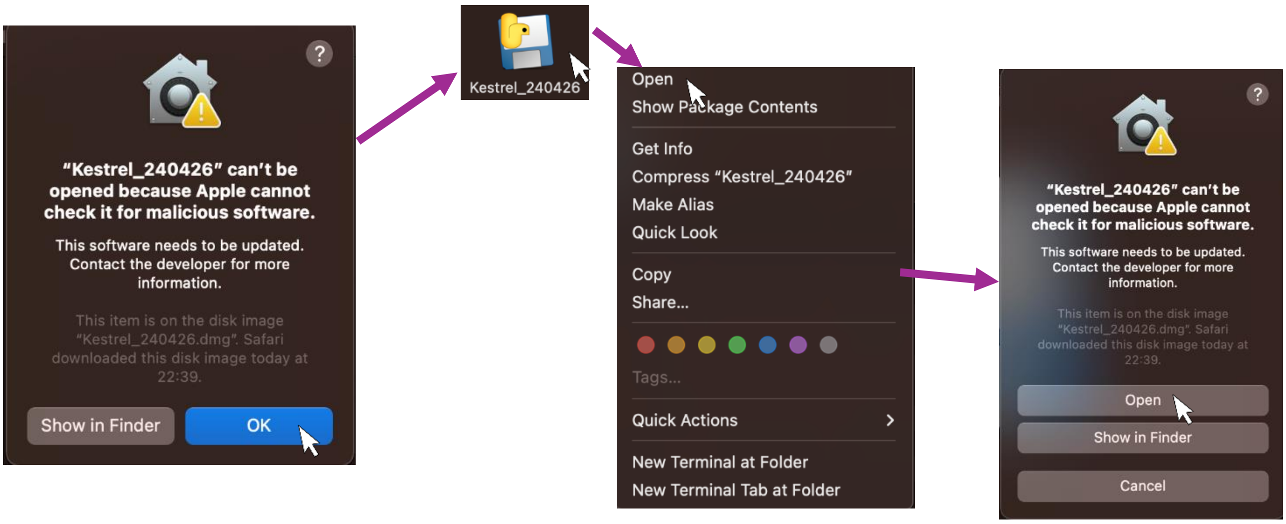



### *Troubleshooting*

i) Depending on system security settings, you may be warned that *Kestrel* cannot be opened.

ii) If this is the case, click "OK" and then right/double-click on the file/thumbnail. This will open a new menu where you should select 'open'.

iii) You will then see the same warning as before but this time there will be an option to "Open". Click this and *Kestrel* will launch.




You are now ready to use *Kestrel*. For new users, see pages 6-12 for a 'quick start' guide which walks through a worked example. The remainder of this guide contains detailed documentation on each calculation and option within *Kestrel*.

## 1.1 Quick Start Guide

**In this tutorial, we will import a set of xyz coordinates into *Kestrel*, parameterise the ligand field with a set of *e* values and display the results of the ligand field calculation.**

**For this tutorial, we will use the example of $[Cu(py)_4]^{2+}$ - a square planar $Cu^{2+}$ complex. This relatively simple system has well characterised experimental data allowing *Kestrel*'s results to be validated.**

Upon launching *Kestrel*, you will see the screen in figure 1. By navigating to the 'File' menu, previous sessions can be loaded as .kes files or a new molecule created by importing a *.xyz* file. For this tutorial, we will be importing an .xyz file which can be generated by downloading "*cupy4.xyz*" from the bottom of our website's documentation page.

Alternatively, copy the coordinates on page 13 into a text editor and save with the '*.xyz*' extension as detailed on page 37

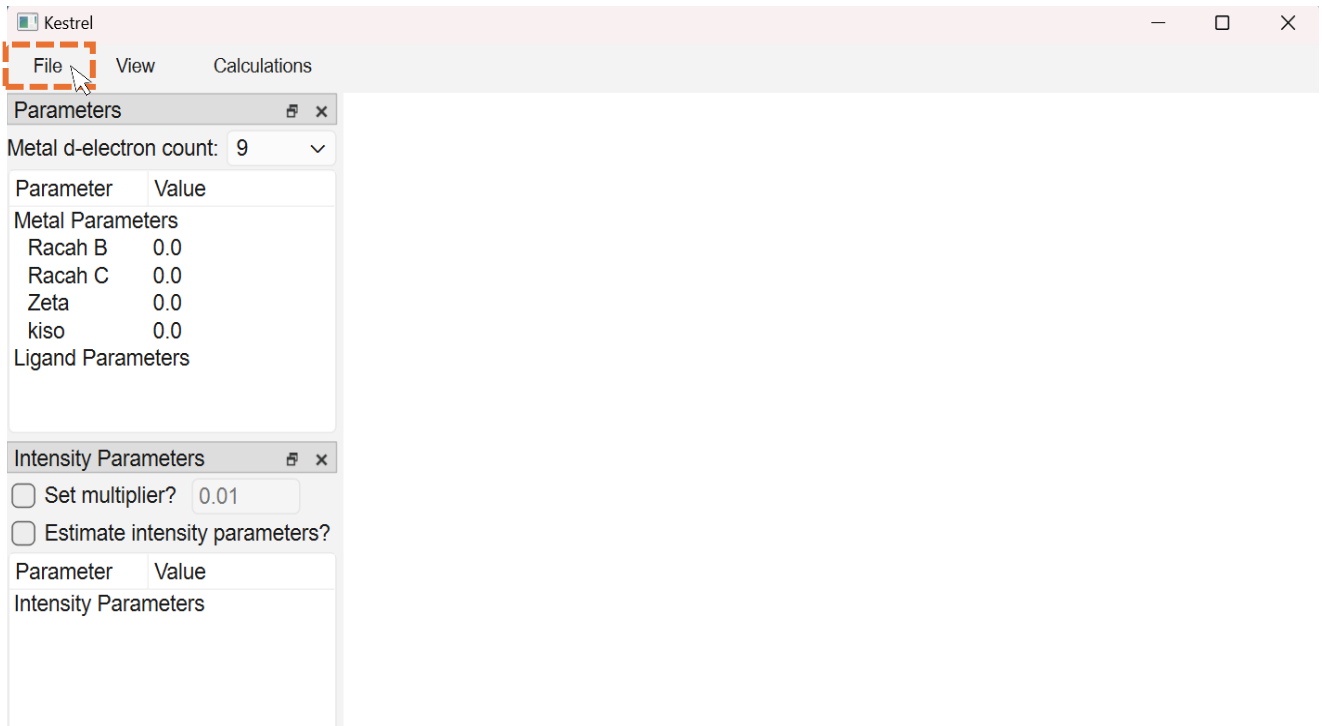



*Figure 1: Appearance of Kestrel's GUI upon launching*

### 1.1.1 Loading a set of .xyz coordinates

In *Kestrel*, click on ‘File’, then ‘Import .xyz’. Navigate to a .xyz file, in this case ‘cupy4.xyz’, and select ‘open’. A window similar to that in figure 2 will then appear.

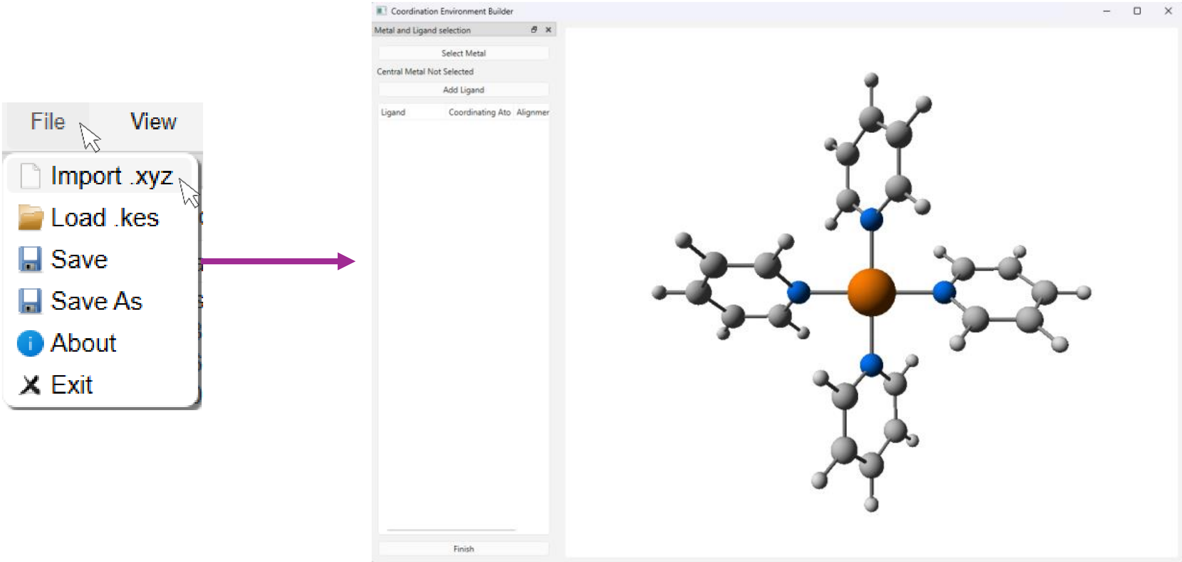
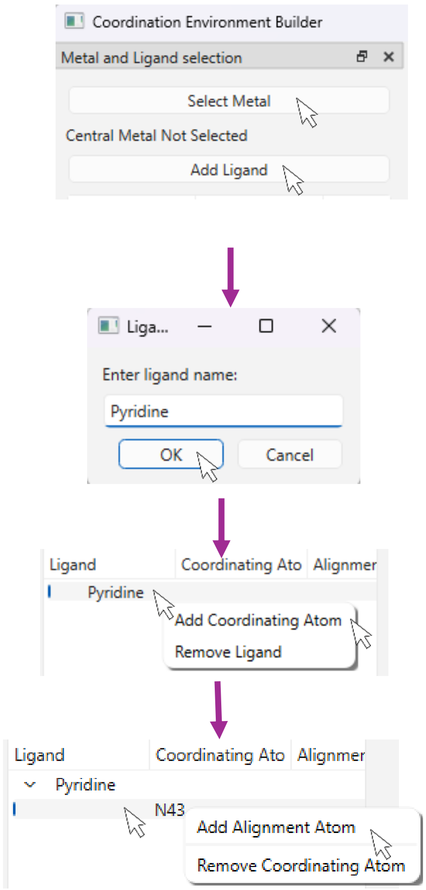



*Figure 2: Appearance of ‘coordination environment builder’ screen*

1. First, assign the metal centre by clicking on ‘Select Metal’ and double-click on the metal atom. The assigned metal will then be shown beneath the ‘Select Metal’ box.
2. Define each ligand type in the complex by clicking on ‘Add Ligand’. Give it a suitable name and select ‘OK’. Here we only have one ligand type, pyridine, but multiple ligands can be defined.
3. Assign the coordinating atoms for each ligand type by right-clicking on the ligand’s name which will open a menu. Select ‘Add Coordinating Atom’ and double click on a coordinating atom. For ligands of the same type multiple coordinating atoms can be assigned.
4. For aromatic rings such as pyridine, it is useful to specify the ligand’s x/y orientation by right-clicking on a listed coordinating atom which will open a menu. Choose ‘Add Alignment Atom’ and double-click on the atom the x/y direction is to be aligned with. This ensures the ligand’s π system is correctly oriented.



Once finished, you should see the same assignments as in figure 3. If you make a mistake, atoms and ligands can be removed by right-clicking and choosing the 'remove' option from the opened menu. Click 'Finish' once all atoms are assigned to close the window.

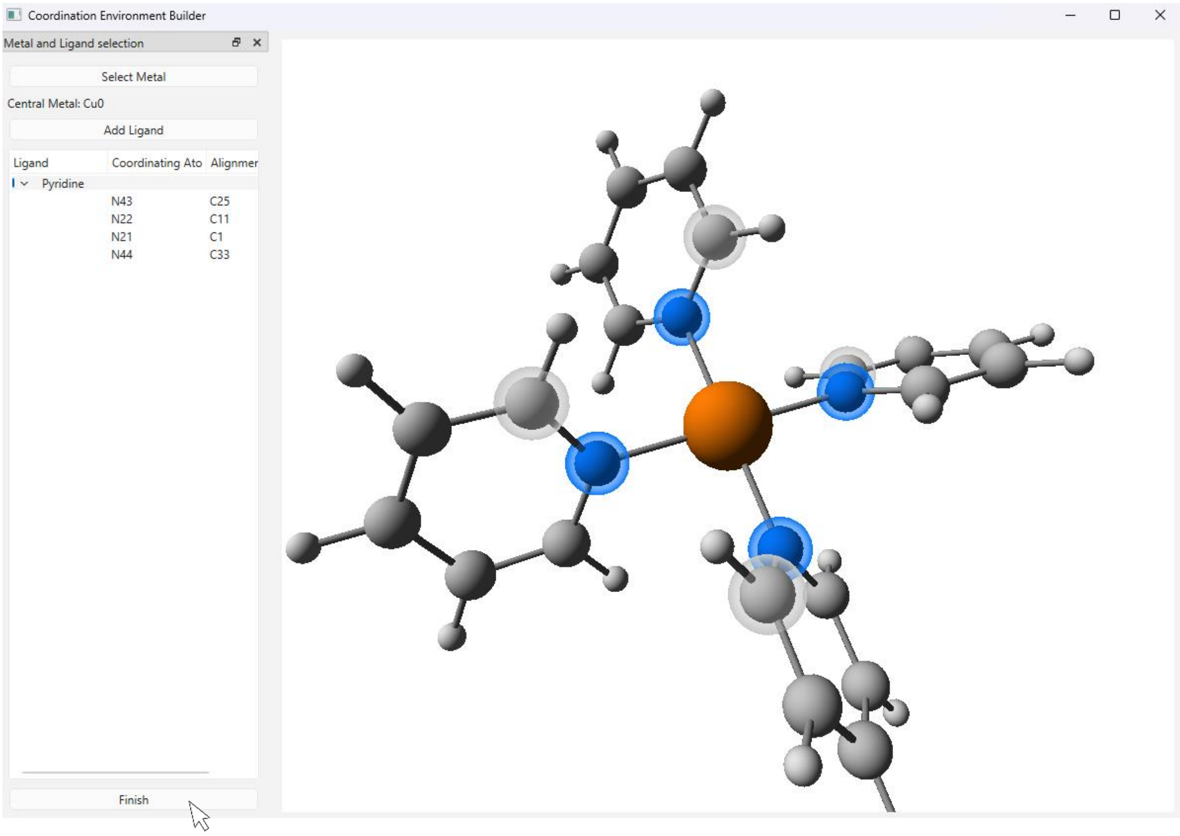



*Figure 3: Coordination environment builder with ligand atoms assigned. The coordinating (blue) and orientation atoms (grey) belonging to the pyridine ligand are highlighted.*

### 1.1.2 Assigning Parameter values

The ‘Parameters’ window on the left of the GUI is where the parameters required for a ligand field calculation are entered. First specify the d-electron count at the top of the panel. The ‘Parameters’ window is then divided into sections for entering metal and ligand parameters. Parameter values are entered by simply typing a value in each box. The parameter values to enter for this example are shown in figure 4 (dashed box).

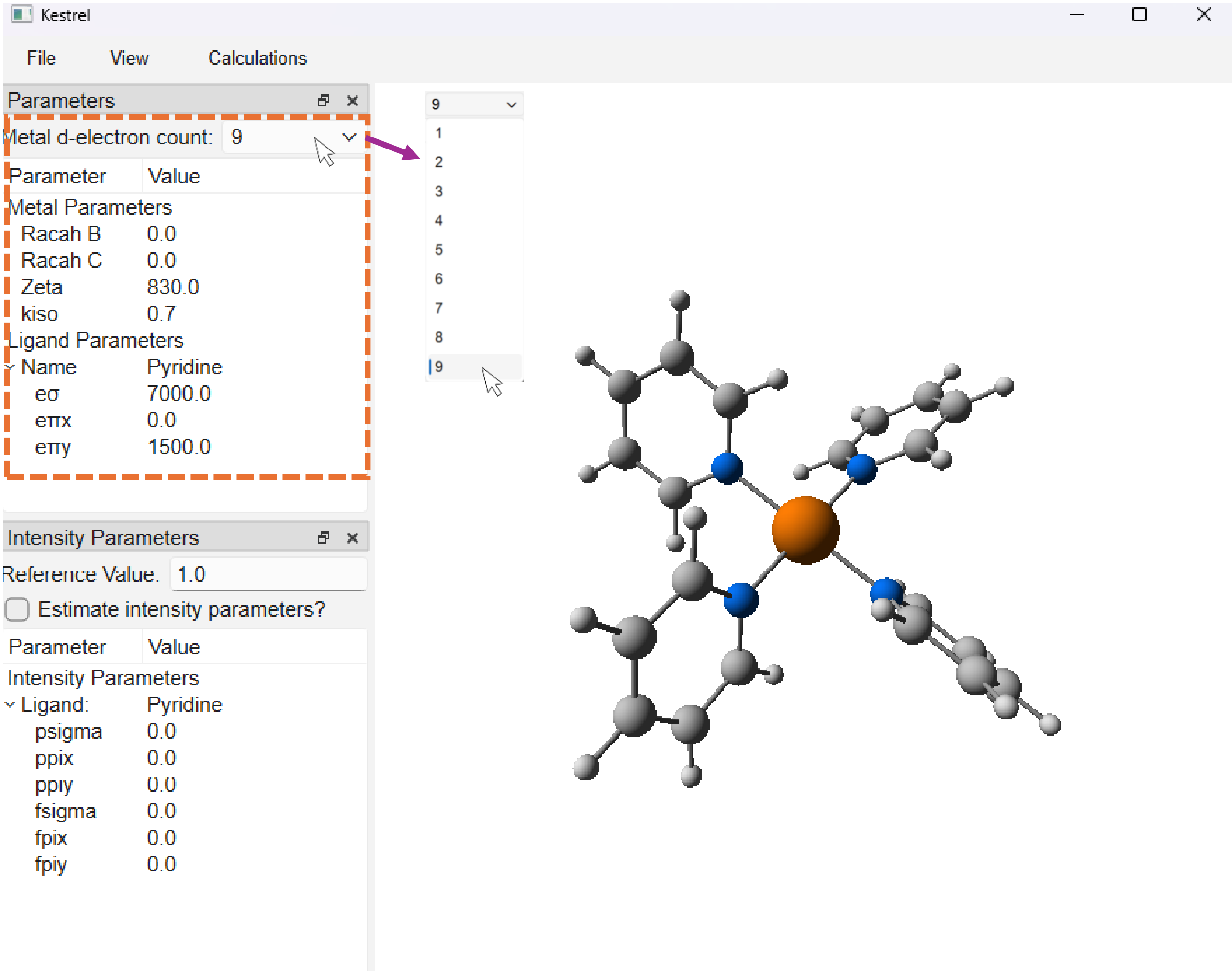



*Figure 4: Main screen of Kestrel’s GUI after importing .xyz coordinates or loading a .kes file along with the parameter values we will use in this tutorial (dashed box).*

### 1.1.3 Further information: A note on parameter values

- Metal parameters parameterise the interelectronic repulsion with Racah B and Racah C values. Spin-orbit coupling (SOC) is parameterised with a Zeta value and orbital reduction factor ($k_{iso}$).
- Ligand parameters describe the strength and type of metal-ligand bond through *e* parameters. The type of *e* value (σ or π) describes the bond and its magnitude the strength of bond.[1]

For this example the metal centre is $Cu^{2+}$, which has a $d^9$ electronic configuration. This can be treated analogously to $d^1$, meaning that Racah B and C can be set to zero to represent no interelectronic repulsion since only a single electron (hole) is present. For SOC, we will use the free-ion value for $Cu^{2+}$ of 830 $cm^{-1}$ and an orbital reduction factor of 0.7. For the ligand parameters, we will use an $e_\sigma$ value of 7000 $cm^{-1}$ and $e_{\pi y}$ value of 1500 $cm^{-1}$. These values have been determined by fitting to experimental Electron Paramagnetic Resonance (EPR) data.[2]

*Kestrel* also has an option to estimate initial ligand parameters by right-clicking on the ligand name and choosing 'Initialise Parameters' (figure 5). The ligand type closest to the one you are parameterising should then be chosen.

Further information about each parameter type and approximate values for each are given in table 1 on page 20.

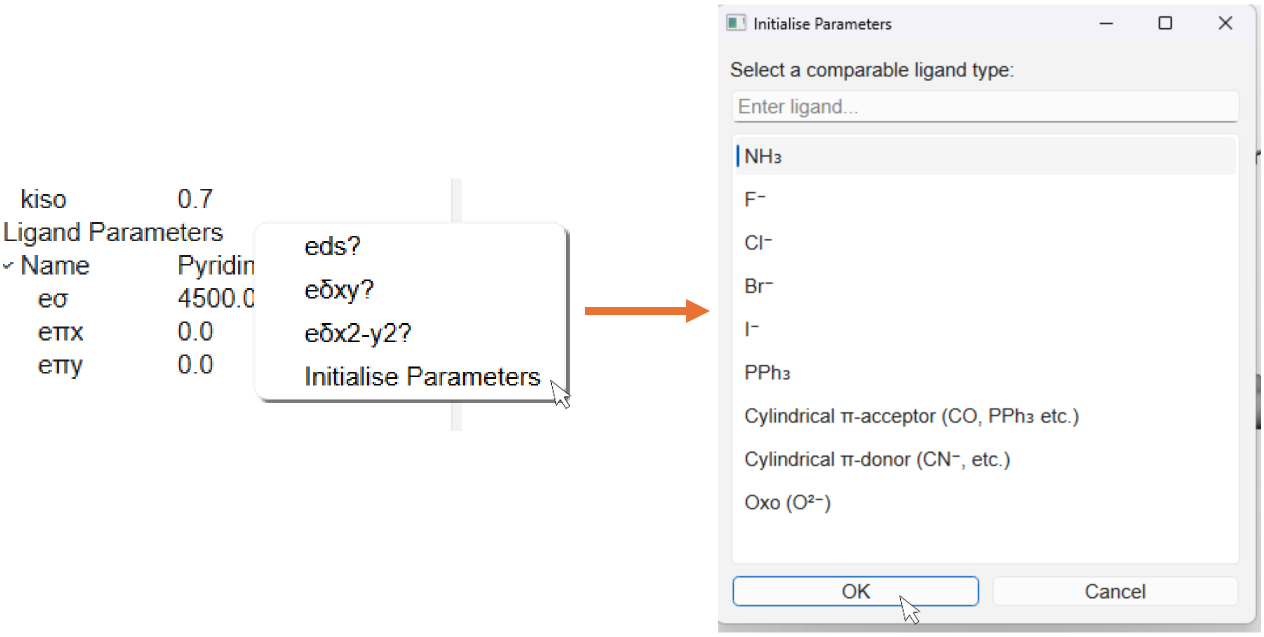



*Figure 5: Initialise parameters dialog which is opened after selecting 'Initialise Parameters'. The ligand type that is most comparable to that in the complex should then be chosen to give a set of starting parameters.*

### 1.1.4 Viewing calculation results

To view the results of calculations, click on the 'Calculations' option in the top menu where initially, only the Ligand Field Matrix and Eigenfunctions options will be enabled. Clicking on an option will then open a window displaying the results of that calculation.

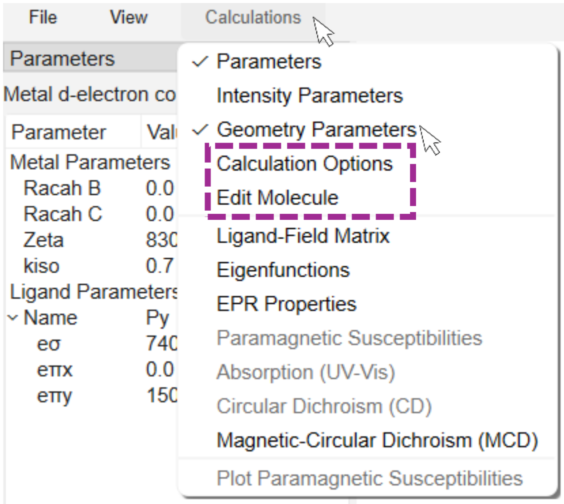



*Figure 6: Calculations menu with options to configure a calculation and view its results*

Additional calculations can be enabled by clicking on 'Calculation Options' which will open a new window. For this example, we will enable EPR calculations by simply checking the 'Enable EPR Calculations' and 'Enable Hyperfine Calculations' boxes.

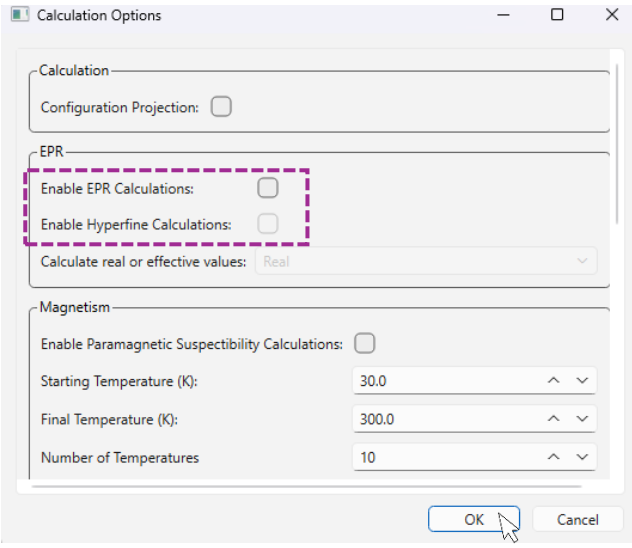



*Figure 7: EPR calculation options within the Calculation Options dialog box*

Figure 8 shows the results of our calculation (some windows have been moved/resized so that all data are visible):

1) EPR g factors and hyperfine values. Experimental g factors are: 2.051, 2.051, 2.262 and hyperfine values are: 48 MHz, 48 MHz, -563 MHz.[2]
2) Ligand field matrix and the energy and composition of each d-orbital. Here we can see the expected square planar splitting of $d_{yz}/d_{xz} < d_{xy} < d_{z^2} < d_{x2-y2}$ due to the lack of ligands in the z-direction.
3) Many-electron eigenfunctions. The energy (and multiplicity) of each electronic state is displayed. The electronic configuration can also be enabled by checking the box 'Configuration Projection'.

Finally, change a parameter value (i.e. increase eσ by 1000 cm$^{-1}$) – the results will update in real-time.

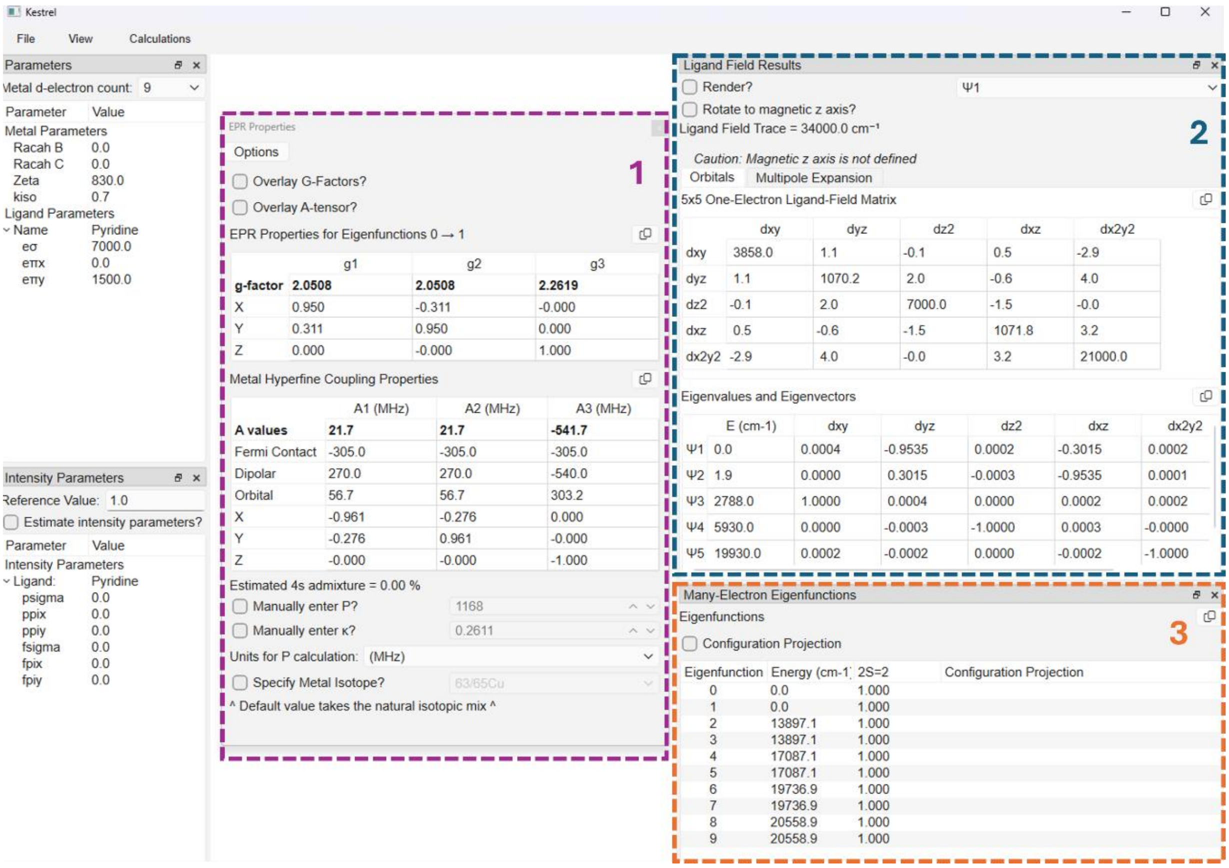



*Figure 8: Results of EPR, ligand field and many-electron eigenfunction calculations.*

To save the calculation, open 'File' and choose 'Save As'. This then creates a .kes file which stores the current parameter values and molecular geometry.

**A .kes file containing the results in figure 8 is available on our website.**

**You are now ready to run calculations on any set of xyz coordinates. The remainder of this document contains a detailed description of each feature and calculation but these details are not essential for successful use of *Kestrel*.**

### *[CuPy4]2+ .xyz coordinates*

Cu 0.00000000 0.000000000 0.000000000
C -0.695174739 2.703797661 -0.935667100
H -1.252849847 2.112386426 -1.661786403
C 0.696234162 2.704292804 0.934554740
H 1.252748139 2.113413780 1.661978708
C 0.710592471 4.094287803 0.969873234
H 1.274833194 4.606919765 1.748082093
C 0.002238170 4.803410619 -0.002494345
H 0.003157740 5.893671184 -0.003448466
C -0.707342372 4.093765099 -0.973548961
H -1.270865502 4.605843838 -1.752638327
C -2.705163278 0.695714907 0.934039059
H -2.114604722 1.251200921 1.662510065
C -4.095195580 0.710639422 0.968342019
H -4.608161030 1.274604700 1.746531287
C -4.803845710 0.003299507 -0.005089467
H -5.894103231 0.004604114 -0.006879539
C -4.093707142 -0.705799809 -0.976140546
H -4.605478125 -1.268587753 -1.755974511
C -2.703782408 -0.694106954 -0.937341313
H -2.112077049 -1.251167681 -1.663681185
N 0.000000000 2.020299387 0.000000000
N -2.020685339 0.000061870 -0.000598595
C 2.703651407 -0.696907651 0.934762350
H 2.112131948 -1.255415688 1.660154620
C 2.704410175 0.696891253 -0.933681483
H 2.113632342 1.254455924 -1.660384628
C 4.094407688 0.711018657 -0.968991246
H 4.607146773 1.276153567 -1.746479720
C 4.803395673 0.001277728 0.002463020
H 5.893656990 0.001972041 0.003404639
C 4.093617228 -0.709406254 0.972614170
H 4.605590090 -1.274043458 1.750967208
C 0.696731329 -2.705326932 -0.932756240
H 1.253364302 -2.114905401 -1.660465621
C 0.711548636 -4.095365066 -0.966887609
H 1.276596203 -4.608479078 -1.744193774
C 0.002708266 -4.803827914 0.005589773
H 0.003881655 -5.894085779 0.007496010
C -0.707716010 -4.093505081 0.975538198
H -1.271697998 -4.605128671 1.754604561
C -0.695796975 -2.703585338 0.936614563
H -1.253849671 -2.111733437 1.662074837
N 2.020290525 -0.000386831 0.000000000
N -0.000206020 -2.020677521 0.000793206

## References

1. M. Gerloch, *Magnetism and ligand-field analysis*, Cambridge University Press, Cambridge, 1st Edition edn., 1983.
2. F. Neese, *The Journal of Chemical Physics*, 2003, **118**, 3939-3948.

## Citation Information

If you use Kestrel in your work, we kindly ask that you provide the following citation:

- G. Nunn, B. S. Grace, M. A. Bates and P. H. Walton, Kestrel [Computer software]. https://github.com/chem-kestrel/Kestrelpy